

# Mika Be

## APPLIED AI | SOFTWARE ENGINEERING

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## EDUCATION

**Rutgers University - New Brunswick** | B.S. Electrical and Computer Engineering, CS minor Expected May 2028

**GPA** 3.838/4.000 | Dean's List, all four completed semesters

**Honors** IEEE-HKN | 2nd Place Best Full Paper, IEEE ISEC 2026 | BCA Computer Science Departmental Award, 1 of 3 recipients

**Coursework** Data Structures; Digital Logic Design; Computer Architecture & Assembly and Linear Systems & Signals (in progress)

## RESEARCH

**Kuiper TTS** | Undergraduate Researcher and Co-author 2025-Present

*Python* | *PyTorch Lightning* | *Piper/VITS* | *ONNX* | *LaTeX* | *GPU training*

- Co-authored "Making Legacy Scientific Papers Accessible by Converting Equations into Spoken Language," awarded **2nd Place Best Full Paper at IEEE ISEC 2026**; coordinated manuscript and reproducibility work for the 18-person team.
- Wrote the narration rules that let a listener follow an equation:  $dq/dt$  is read as "the derivative of q with respect to t," and fractions, partial derivatives and chained equalities get explicit grouping and pauses.
- Recorded 200 training sentences in my own voice with a custom recording tool, then rebuilt a broken Piper/VITS training stack into a reproducible GPU pipeline to fine-tune a voice for reading science, exported to ONNX for offline playback.

## PROJECTS

**Room of Days** | Creator and Solo Developer 2025-Present

*Flutter* | *Dart* | *Swift (WidgetKit)* | *Firebase* | *Cloud Functions*

- Started and built a life-RPG habit app that turns everyday tasks into Quests: finishing one earns XP, grows stats like Mind and Home, and unlocks furnishings for a room of your own. **10+ people use it.**
- Shipped **49 builds** on my own, covering product and visual design, a custom score, sound and haptics, and an offline-first Firebase backend with backups, optional sync and account deletion.
- Added .ics class-schedule import with reminders and an iPhone Home Screen widget; 1,200+ automated tests.

**Room Notes** | Creator and Solo Developer, in use in my own classes Aug 2026-Present

*Flutter* | *Dart* | *Firebase* | *Drift/SQLite* | *Flutter iOS engine* | *Codemagic CI*

- Building the notebook I wanted for engineering classes: handwritten on an iPad, capture the board with a phone, type equations and code on a laptop, and find all of it on the same page with the same layout on every device.
- Patched Flutter's iOS engine** to read Apple Pencil's coalesced touch samples for smoother handwriting, and set up CI that compiles the custom engine and ships it to TestFlight.
- Designed local-first sync that saves each edit with its pending upload in one transaction, so notes survive offline use and restarts; 1,700+ automated tests.

**Knowsy Breadboard Lab** | Technical Lead, three-person team 2025-Present

*JavaScript* | *circuit simulation* | *Verilog* | *automated testing*

- Built a browser breadboard for Rutgers Digital Logic and Principles of EE II labs: students wire 16 real 74LS-series chips through 8 lab modules, from a 4-bit ALU to a traffic-light controller and an active bandpass filter.
- Wrote both simulators from scratch with **no math libraries**: a digital engine that catches floating inputs and shorted outputs, and a modified-nodal-analysis solver that drives an oscilloscope and a Bode plotter.
- Added export of any bench build to Verilog that runs in Icarus Verilog; 280+ automated tests.

## EXPERIENCE

**this.is** | Lead Software Developer, Internship Jul 2025-Aug 2026

*React* | *TypeScript* | *Firebase* | *Cloud Functions*

- Led development of a social discovery platform as its primary engineer: search, profiles, saved lists and sharing.
- Prototyped embedding-based search with query expansion, and wrote a load-testing harness that reports p50/p95/p99 latency and per-query cost for 300 simulated users.
- Drove a product reset toward private groups of 2 to 6 friends, showing the places where their saved tastes already overlap.

**Schieber Research** | Software Engineering Intern 2023-2024

- Built and documented a React/Vite research prototype backed by Appwrite from the team's study requirements.

**Starbucks** | Barista, part-time alongside full-time coursework 2023-2025

## TECHNICAL SKILLS

**Languages** Python, Dart, JavaScript/TypeScript, Java, C/C++, SQL, Verilog, HTML/CSS

**AI and research** PyTorch Lightning, Piper/VITS, ONNX, embeddings, GPU training, speech datasets

**Software** Flutter, Firebase/Firestore, Cloud Functions, React, Drift/SQLite, Git, CI/CD (Codemagic), TestFlight

**Hardware** Digital logic, 74LS-series ICs, breadboarding, ESP32, LTspice, oscilloscopes